Sp. Tp. Human Computer Interaction
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Hue Hunter

1. Introduction:

Hue Hunter is a color-matching game made using HTML, CSS, and JavaScript. The user is presented with a target color, and their job is to match this color using three sliders representing the Red, Green, and Blue values. In each round, a score is calculated, which is then translated to coins that can be used in the Wardrobe to create a pixel art avatar. This paper will analyze the functionality and use of HCI concepts in the application.

2. Code Overview:

This is a single-page web application consisting of one HTML document with integrated CSS and JavaScript. It is structured around five different modules including the Color Mixing module, the Hint System, the Odd One Out game, the Wardrobe module, and the Tutorial module. All of the state information is saved using the browser's local storage.

3. Screens and Functionality:

Start Screen

At the very beginning of the process, there is the title, the tag line, and the image of the player's pal. The key action to be performed by the user at that point is Start Playing, with the links to the wardrobe and to help being presented as secondary elements. New users will be provided with an option to select their character – either a girl or a boy.

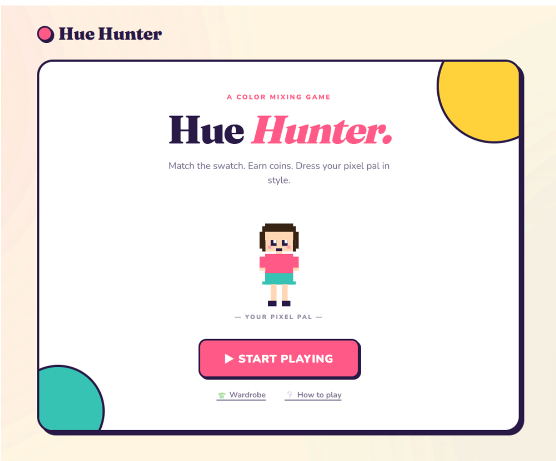

Fig.1 Start Screen

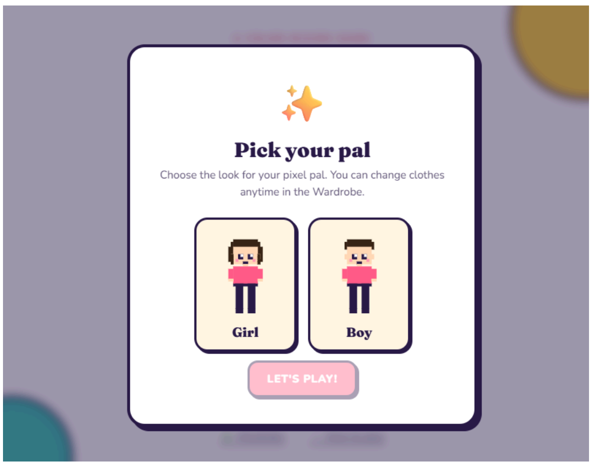

Fig.2 Character Selection

HCI Principles Applied

•**Familiarity (Dix et al., Learnability):** The layout follows the universally familiar pattern of title → tagline → primary button, matching user expectations for the first screen of any game.

Game Screen

In the Game Screen, there are two swatch cards, namely Target color and Your Mix, respectively positioned to the left and right of the screen. Three sliders are present below the swatch cards, and each slider is labeled using color coding. A constant header displays the Level, Score, Coins, mini character, and a question mark button. In the target swatch, there are three pairs of double underscores ("__") that can be clicked to get hints; hints are represented by a yellow pill, while "+ Earn" leads to Odd-One-Out game.

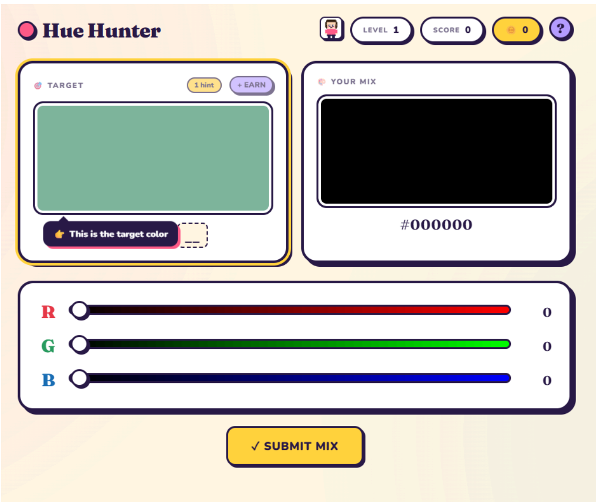

Fig.3 Game Screen with the first instructions on how to use

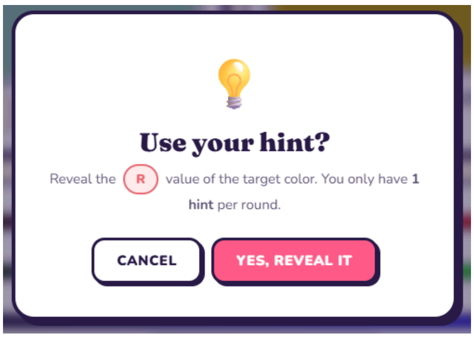

Fig.4 Hint usage pop-up

HCI Principles Applied

•**Natural Mappings by Norman (1986):** The sliders have colors that reflect the effects that they control, while moving them causes a continuous effect in the mix.

•**Fitts' Law (1954):** Big objects lead to shorter times in making movements, hence, the slider thumbs are sized at 28 pixels and the Submit Mix button is given generous padding.

•**Principle of Contiguity (Mayer, 2009):** The slider's channel name, track, and numeric value appear on the same row in order to allow for the proximity of the elements that the user needs.

•**Error Prevention (Rule 5 of Shneiderman; Nielsen):** When selecting any hint pair, a confirmation message appears to protect against the limited hint being wasted.

Odd-One-Out Mini-Game

The mini-game enables the player to get additional hints. The game presents a 2×2 grid, containing four squares. Three are alike while the fourth one differs a bit from the other three. The player gets five seconds to find which one is different. The player's progress is tracked by a timer bar that drains away in real-time mode. There is an increase in difficulty level at each level.

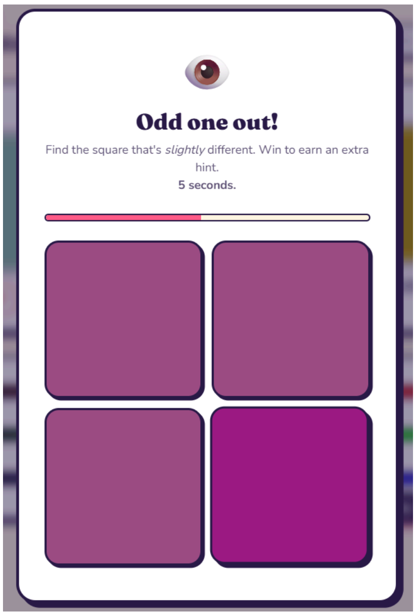

Fig.5 Odd one out mini game

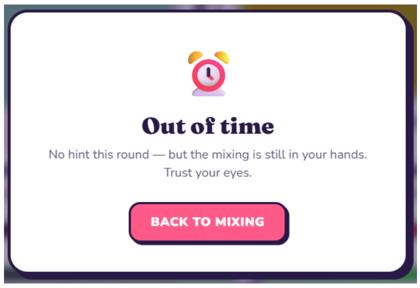

Fig.6 Out-of-time pop-up for mini game

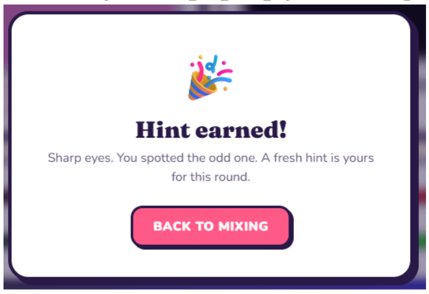

Fig.7 Hint earned for mini game

HCI Principles Applied

•**Signal (Mayer, 2009):** The timer bar displays elapsed time, while a change to red color below 30% alerts users to the need for urgency without the need to monitor time by the second.

•**Informative Feedback/Closure (Shneiderman Rules 3 & 4):** A right answer generates a glowing teal effect animation, an incorrect answer shakes the tile, and running out of time triggers an out-of-time message.

Result Screen

Once a mix is submitted, the player gets an indication of how accurate they were in terms of a big percentage accompanied by an appropriate emoji and a slogan. There is a side-by-side display of the target alongside the mix submitted by the user. The coin burst animation reveals how many coins have been earned. There are three buttons for moving forward.

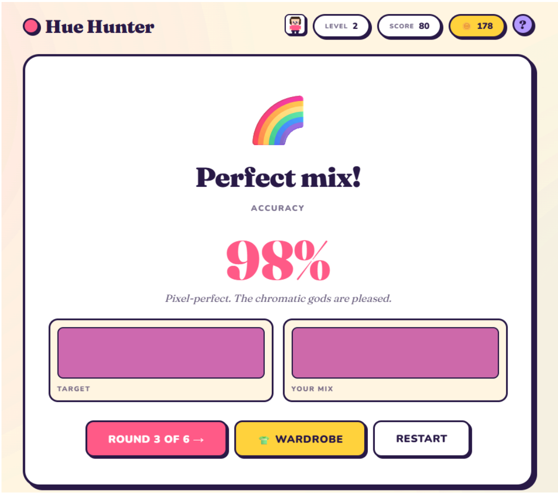

Fig.8 Result Screen with accuracy and coin burst

HCI Principles Applied

•**Synthesizability (Dix et al., Learnability):** Placing the target and user's mix side by side lets the player directly assess the effect of their past actions.

Wardrobe Screen

Wardrobe allows the player to purchase cosmetic items using coins. There is a preview of the character on the left while there are three tabs on the right that categorize the items. The item card includes the thumbnail of the item, its name, and either its cost or status. Locked items have the Purchase Confirmation modal that disables the purchase button when necessary.

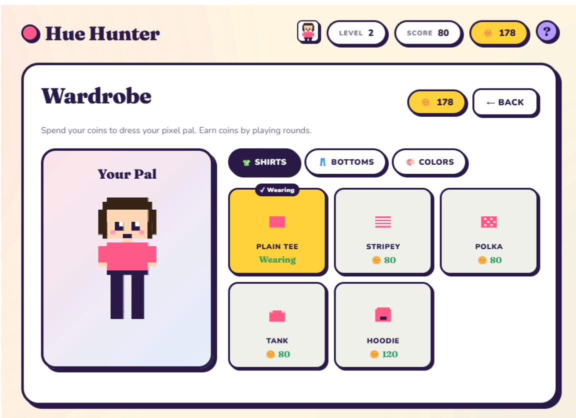

Fig.9 Wardrobe Screen

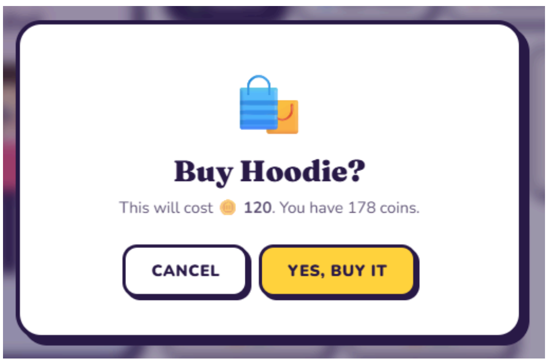

Fig.10 Purchase Confirmation

HCI Principles Applied

•Hick-Hyman Law (Hick, 1952; Hyman, 1953): Instead of placing all items into one matrix, The Wardrobe divides the selection process among three tabs, which helps to decrease n and thus cut down on the decision-making time.

•Error Prevention and Ease of Undo (Shneiderman Rules 5 & 6): The purchase process involves a confirmation dialog, which prevents the user from making erroneous purchases. Furthermore, The Buy button becomes inactive if there are not enough coins. Equipped gear can always be replaced by tapping on it once.

•Proximity (CRAP): In The Wardrobe, each item card displays all its preview, label, and price right inside one border-lined box that stands apart from other cards via whitespace.

Tutorial and Help

During the first round only, a step-by-step tutorial walks the player through each interaction. Each step adds a glowing outline and an arrow tooltip pointing to the relevant element. Once completed, the tutorial never appears again. A persistent help (?) button opens a five-step How to Play guide with an expandable explanation of RGB color mixing.

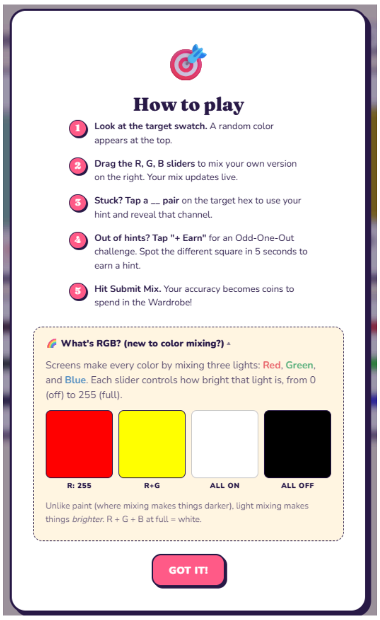

Fig.11 Help Screen

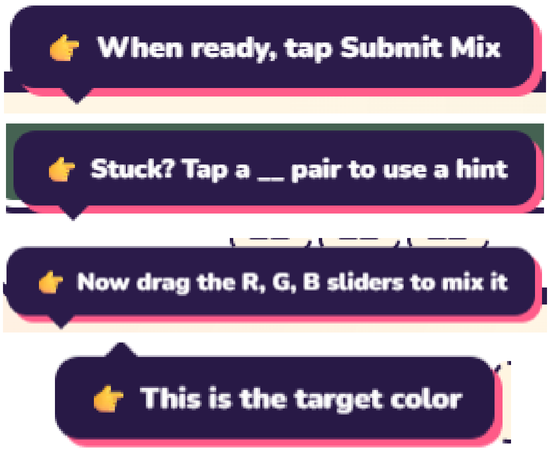

Fig.12 Tutorial pop-ups when playing for the first time

HCI Principles Applied

•Signaling (Mayer, 2009): Each tutorial step adds a glowing yellow outline and an arrow-tipped tooltip, drawing attention to exactly where the user needs to act next.

•Pre-training (Mayer, 2009): The Help modal includes an expandable "What's RGB?" section introducing the underlying concept before the user needs to apply it.

•**Predictability (Dix et al., Learnability):** Because the tutorial shows the user the immediate effect of each step before they perform it, the user develops a clear expectation of what will happen when they act.

4. System-Wide HCI Principles:

Beyond the screen-specific principles discussed above, several principles apply at the level of the system as a whole.

a. CRAP

Contrast: The dark navy text on the creamy background maintains readability; the buttons are colored crimson with white text for visibility; R/G/B text tags and hex values are colored using their respective color codes to build the cognitive model.

Repetition: All cards, modals, and buttons employ the same border, shadow, and border-radius styles. There is just one layout design used by all seven confirmations, allowing the user to memorize the modal layout only once.

Alignment and Proximity: Slider rows have label, track, and value aligned in one line; the header section groups stats on the right side and identity on the left side, forming two distinct clusters on all screens.

b. Consistency and Robustness (Shneiderman; Dix et al.)

The color of primary actions should always be crimson pink, that of secondary actions white, while yellow is used for highlighting confirmations and rewards. In modals, the Cancel button should always appear on the left side while the Confirm button on the right side.

Observability is achieved with an ongoing header that will always display to the user the current mix of coins, hints, and levels. Recoverability happens on three separate levels: every confirmation modal will have a cancel option; the wardrobe has equipped clothing items that the user can switch or remove at any time and without charge, and the user can return to their start screen from any other screen.

5. Conclusion:

Hue Hunter is an interactive and functioning game that allows players to mix colors, with a clearly established feedback system and customizable character. The prototype shows how the application of CRAP design principles, Shneiderman's golden rules, usability principles by Dix et al., multimedia learning principles by Mayer, and Norman’s mappings and Fitts’ law create a refined user interface.

These principles interact with one another; a tutorial (Mayer) is successful because its controls use natural mapping (Norman) and an error preventing confirmation dialog (Shneiderman) is coherent because it follows the same template as every other modal (Repetition).. The result is that a small, finished prototype illustrates how HCI principles can create a more polished and understandable experience, even for what is considered to be a simple game.

6. How to Run?

Open HueHunter.html in any modern web browser. No installation, build step, or server is required. Progress is saved automatically to the browser's localStorage and persists between sessions on the same device.

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